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# Edge Computing vs Cloud Computing

## Choosing the Right Computing Paradigm

A comparison of architectures, applications, and industry adoption scenarios




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**Course name : CLOUD COMPUTING & BIG DATA ANALYTICS**

# Two ways to process data

## ☁ CLOUD COMPUTING

- Processes and stores data in remote data centers
- Resources are accessed over the internet
- Scales up for large workloads
- Centralized management and analytics

**Examples: AWS • Microsoft Azure • Google Cloud**

## ⚡ EDGE COMPUTING

- Processes data near where it is generated
- Reduces dependence on distant servers
- Enables faster, low-latency responses
- Useful for IoT and real-time systems

**Examples: IoT gateways • smart cameras • factory devices**

**CLOUD = COMPUTE AT SCALE | EDGE = COMPUTE NEAR THE SOURCE**

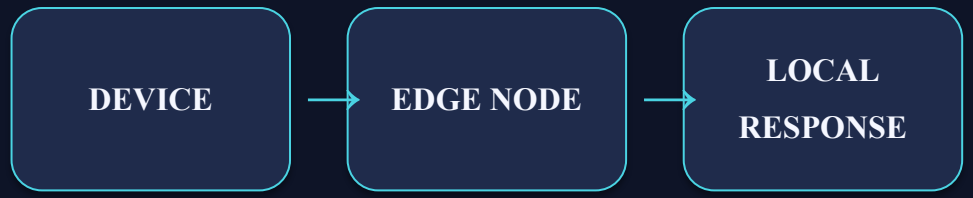
# Where does the processing happen?

## CLOUD ARCHITECTURE




Data travels to a centralized server for processing.

## EDGE ARCHITECTURE




Data is processed nearby for quick action.

| FACTOR | CLOUD | EDGE |
|---|---|---|
| **Latency** | Often higher | Usually lower |
| **Connectivity** | Internet often needed | Can work locally |
| **Best fit** | Scale & heavy analytics | Fast, local decisions |

# Different strengths, different workloads

## CLOUD COMPUTING

- Backup and large-scale storage
- Web and mobile applications
- Big-data analytics
- AI/ML model training
- Streaming and collaboration platforms

## EDGE COMPUTING

- Autonomous and connected vehicles
- Smart surveillance cameras
- Industrial IoT and robotics
- Healthcare monitoring devices
- Smart traffic systems and gaming

**Example: an edge-enabled camera can detect an event locally and alert immediately.**

# How industries combine edge and cloud

## MANUFACTURING

EDGE

Monitor equipment in real time; flag anomalies near the machine.

## AUTOMOTIVE

EDGE

Support fast in-vehicle decisions; use cloud for fleet insights.

## HEALTHCARE

HYBRID

React to local sensor alerts; analyze trends across long-term records.

## RETAIL & SMART CITIES

HYBRID

Use edge for checkout/traffic responses and cloud for broader analytics.

**Adoption depends on response time, connectivity, data volume, privacy and cost.**

# Which paradigm should you choose?

## CHOOSE CLOUD WHEN…

- You need massive storage or compute
- Workloads require complex analytics
- Centralized access and management matter
- Some network delay is acceptable

## CHOOSE EDGE WHEN…

- A near-instant response is essential
- Devices generate data continuously
- Connectivity is limited or unreliable
- Local processing supports privacy

## BEST OF BOTH: HYBRID EDGE + CLOUD

Edge handles immediate decisions • Cloud handles large-scale storage and analytics

# Edge and cloud are complementary

- Cloud provides scalability, centralized processing and large storage.
- Edge delivers low latency, local processing and rapid response.
- Many real-world systems use both paradigms together.
- Choose based on latency, data volume, connectivity, security and cost.

THE TAKEAWAY

**“Process data at the edge when speed matters; use the cloud when scale matters.”**

## THANK YOU!

# THANK YOU !!!